number = float(input("Please enter number")). We can also use + to concatenate string in Python. print("20" + "20" + "20") → 202020. We can use f-strings to format our print. print(f"The number is: {number:>+10.2f}") → The number is +20.00. This means the number 20 will be right aligned in 10 spaces, have a sign and 2 decimal places. WK1

A list stores multiple items in one variable. Lists are mutable. Indexing gets one item and starts at 0. Slicing gets several items. .append() adds an item to the end. remove() removes the first matching item. List comprehension is a shorter way to create a list using a loop you can also use a condition. Tuples store multiple items but cannot be changed after creation. Use () to create a tuple. Dictionary store information as key-value pairs. Add or update a ~~key~~ value using d["key"]= value. We loop over dictionaries using .keys(), .values() and .items(). A set stores unique values only. It removes duplicates. We use open() to read or write files. "w" overwrites a file; "r" reads a file. "a" appends a file. .strip() removes whitespace. .split(",") separates text wherever a comma appears. Use try and except to stop expected errors from crashing your program. List of dictionaries is a common way to store multiple records. with each record represented by a dictionary.

```
colours = ["red", "blue", "green"]
print(colours[0:2]) → red, blue
colours.append("pink")
colours.remove("blue")
squares = [n**2 for n in range(1,4)]
sizes = ("S", "M", "L")
student = {"name": "Shanice", "age": 19}
student["age"] = 20
```

WK4

```
numbers = {2,4,4,6}
print(numbers) → {2,4,6}
with open("records.txt") as f:
    text = f.read()
try:
    age = int("abc")
except ValueError:
    print("Please enter a number")
```